Poster rundown: what goes where

Prepared for Cesar Sierra, MDST EEG project. The attached reference supplies the layout idea: a blue title banner, a compact left methods column and a larger results area. Scientific content and figures here come from your actual dataset and held-out predictions; the reference poster's microscopy images and logos are not reused.

Piece	Content to place
Title / identity	EEG Patterns in Alcoholic and Control Participants. Add approved author names and affiliations; currently marked “Prepared by Cesar Sierra.”
Introduction	Brief EEG definition; recognition task; the “Who?” and “How?” questions.
Dataset	20 people, 600 trials, 64 recorded channels, 256 samples/trial; 10 people/group and 200 trials/condition.
Methods	Cleaning, filtering, feature extraction, participant-separated nested validation.
Model 1 panel	Spatial participant classifier; 85% across three splits; accuracy and confusion figures.
Model 2 panel	Stimulus classifier; 57.78%; accuracy, confusion and descriptive group comparison.
Conclusion / limitations	Original spatial method remains the best tested participant approach. Small cohort, no external validation, unverified onset, uncalibrated probabilities.
References / thanks	UCI dataset, Week 1 slides, analysis logs; MDST project leads. Confirm authorship/acknowledgments before presenting.

Ready-to-paste introduction

Electroencephalography records voltage fluctuations at electrodes on the scalp. We analyzed EEG recorded during a visual recognition task to ask whether signal features distinguish alcoholic and control participants and identify stimulus conditions. We compared participant and condition models using validation that holds out entire participants.

Dataset note

The analyzed file is the original cleaned training subset, not the entire public database. It contains 9,830,400 voltage rows = 600 trials × 64 channels × 256 samples. There are 38,400 trial-channel recordings. The current cleaned export retained all rows and all trials. The full UCI database is larger; do not label these results as a 122-participant evaluation.

Model 1: answer “Who?”

Slide question: Can EEG classify alcoholic versus control participants?

Answer: In this original subset, spatial EEG features supported classification of held-out participant labels with 85% mean balanced accuracy. Scores were 80%, 85%, and 90% under three fixed participant splits. This is an exploratory within-dataset finding; it does not establish clinical diagnostic accuracy on new populations.

Ready-to-paste model methods

We estimated within-trial OAS shrinkage covariance and correlation from 61 named scalp channels after 1–40 Hz filtering. X, Y and nd were excluded from this branch. Symmetric matrix-log features represented relationships between electrodes. Features were aggregated within each participant and condition. Inner validation selected among mean covariance, median covariance, and mean correlation pipelines, each using training-fold imputation, scaling, six-component PCA and balanced logistic regression (C=0.1).

The final saved spatial refit uses mean log-covariance features. Its 1,891 covariance values per condition yield 5,673 participant features before PCA. The reported 85% evaluates the nested selection procedure, not an independent test of the saved model refitted on all participants.

Ready-to-paste results

Mean balanced accuracy was 85.00%; mean macro-F1 was 0.850. Across the three splits, pooled held-out decisions correctly classified 27/30 alcoholic decisions and 24/30 control decisions. These 60 decisions represent the same 20 people evaluated three times, not 60 independent participants. Frequency-specific and geometry models, as well as 18 later spatial adjustments, did not exceed this mean.

Figures to place

participant_accuracy: three split scores with a 50% balanced-class chance line.
participant_confusion: normalized row percentages and pooled decision counts. Label repeated decisions explicitly.
workflow: compact pipeline graphic, optionally placed in the methods column.

What the model does not show

It does not establish a causal effect of alcohol, localize a neural generator, or prove which electrodes are clinically meaningful. Covariance is a signal relationship, not a direct measure of anatomical connectivity. The probabilities are uncalibrated: a classifier output of 0.8 is not a verified 80% clinical risk.

Model 2: answer “How?”

Slide question: Do alcoholic and control brains respond differently to recognition tasks?

Answer: We can decode the stimulus condition above balanced-class chance, but these model results do not establish different neural responses between groups. Overall condition accuracy averaged 57.78%. Group-level predictive performance was nearly equal: 57.67% for alcoholic participants and 57.89% for controls.

Ready-to-paste model methods

Trial-level features describe full-epoch time-domain and spectral patterns plus eight recording-relative 125-ms bins per electrode. A fixed candidate set included the earlier temporal/spectral pipelines and added spatial/temporal options. Inner participant folds selected models; outer participant folds evaluated the three conditions. The saved all-data refit uses 29 features per channel (1,856 across 64 channels), training-fold imputation/scaling, selection of 256 features and balanced logistic regression (C=0.1). Outer folds may select different candidates.

Ready-to-paste results

Balanced accuracy was 59.17%, 57.33%, and 56.83% across the three fixed splits (mean 57.78%; mean macro-F1 0.575). Pooled class recall was 70.7% for S1, 56.5% for S2 match and 46.2% for S2 non-match. Confusion between the second-stimulus conditions was common. The three-class balanced chance reference is 33.33%.

After averaging each participant's performance across seeds, the alcoholic-minus-control accuracy difference was -0.22 percentage points. An exploratory participant bootstrap interval was -7.78 to +6.78 points. It includes no difference. This compares predictive performance, not a formal physiological response effect; no mechanistic or causal claim follows.

Figures and cautions

condition_accuracy: three split scores with the 33.33% chance line.
condition_confusion: 1,800 held-out decisions from 600 trials repeated across three seeds.
group_condition_performance: ten participant dots per group, means and participant bootstrap intervals.
FZ_waveforms: descriptive recording-relative means ± SEM after averaging trials within each participant. These are not confirmed stimulus-locked ERPs; apparent waveform differences are not tested group effects.

Answers to the Week 1 slides

Slide topic / question	Answer
What is EEG? (p. 7)	A recording of voltage changes over time from electrodes on the scalp. Each electrode signal forms a channel; EEG is not a direct picture of individual neurons.
What is cognitive modeling? (p. 8)	Using computational/statistical models to study patterns associated with information processing and task responses. Here, signal features predict dataset groups or conditions; this is predictive modeling, not a validated mechanistic cognitive model.
What are electrodes, channels, trials and epochs? (p. 9)	Electrode: scalp sensor. Channel: its recorded signal. Trial: one experimental repetition. Epoch: the signal segment analyzed for that trial. Sample: one voltage measurement. Subject: one participant.
How is this dataset organized? (pp. 9–10)	One row per participant–condition–trial–channel–sample, with voltage in microvolts. A trial has 64 × 256 rows; the analysis epoch lasts about one second.
What did participants do? (p. 10)	Viewed objects in S1 single-stimulus, S2 matched, or S2 non-matched conditions. Match/non-match refers to whether the second object matches the first.
What are we building? (p. 6)	Two models: participant group classification and stimulus-condition prediction, followed by a descriptive comparison of prediction performance by group.
How do the tools fit? (p. 11)	NumPy: arrays/matrix algebra. Pandas: tables and identities. SciPy: filters/spectral features. Scikit-learn: pipelines and validation. Matplotlib: these charts. Streamlit is a possible future dashboard; it was not built in this poster task.

Pandas exploration answers (pp. 12–13)

head(): previews sample rows. shape: the analyzed CSV has 9,830,400 rows and 9 columns. columns: subject_id, group, condition, trial, channel, sample, voltage_uv, source_file, parse_error.
info(): inspect dtypes and nonmissing counts; exact dtypes depend on CSV import settings. describe(): summarizes numeric columns, but dataset-wide voltage means hide participant/channel/time structure. unique(): verify 20 participant IDs, two groups, three conditions and 64 channel labels. isnull().sum(): count missing values by column. An empty parse_error means no recorded parse error; it is not a missing voltage.

For a first EEG plot, filter to one participant, one trial and one channel; sort by sample and plot voltage_uv against sample/256. For a task comparison, average trials within participants before averaging participants so people, rather than millions of voltage rows, define the comparison. Administrative survey prompts and the personal theme-song icebreaker are not scientific poster questions.

Figure captions, validation and references

Reusable figure captions

1. Validation accuracy. Balanced accuracy under five-fold outer participant validation, with three-fold inner selection, repeated for seeds 183–185. Seeds reuse the same 20 participants. Dashed lines show balanced-class chance, not a significance test.

2. Participant confusion. Row-normalized confusion over 60 held-out classification decisions from 20 participants across three seeds. Alcoholic recall: 90%; control recall: 80%. Repeated decisions are not independent participants.

3. Condition confusion. Row-normalized confusion over 1,800 held-out decisions from 600 trials across three seeds. All three conditions have equal support. The S2 conditions are more difficult to distinguish than S1.

4. Condition performance by group. Each dot is one participant's accuracy averaged across seeds. Diamonds show group means and exploratory 95% bootstrap intervals obtained by resampling participants within each group. The comparison does not test a neural mechanism.

5. FZ recording-relative traces. Filtered FZ voltage, averaged first over trials within each participant/condition and then across ten participants per group; shading is ±1 SEM across participant means. This is descriptive and is not a verified stimulus-locked ERP or a hypothesis test.

Why the calculations are defensible

All classifiers, imputers, scalers, feature selectors and PCA transforms are fitted on training participants. Inner validation never includes outer-held-out participants. Each seed predicts every original participant/trial exactly once per evaluated procedure. The poster metrics are independently recomputed from the saved predictions. Raw waveform illustration uses actual recorded voltages and does not enter model selection. Mean accuracies are not averaged training scores.

What remains to validate

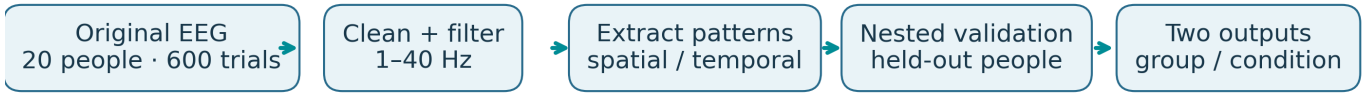
Only 20 independent people were evaluated; millions of samples do not increase that participant count. These people have been used in repeated development experiments. External participants are needed to establish generalization. Confirm acquisition/event metadata before ERP or baseline-window analysis. A formal group-by-condition physiological analysis would require prespecified measures, participant-level inference and appropriate multiplicity control.

References and acknowledgments

1. UCI Machine Learning Repository. EEG Database, dataset 121. https://archive.ics.uci.edu/dataset/121/eeg+database (accessed 11 October 2026). UCI documentation describes 64 channels at 256 Hz and the recognition conditions.
2. Michigan Data Science Team. EEG Cognitive Modeling: Week 1, Kickoff & Brain Basics. Supplied slide deck, especially pp. 6–13.
3. Original analysis artifacts: all_oof_predictions.csv, participant_splits.json, inner_selection_scores.csv, metrics.json and spatial quality logs. Figure code and plotting data are included.
Acknowledgments: MDST project leads Manishika Balamurugan, Panna Akter and Swetha Chimalakonda; public dataset contributors. Confirm approved author/team names before presenting.

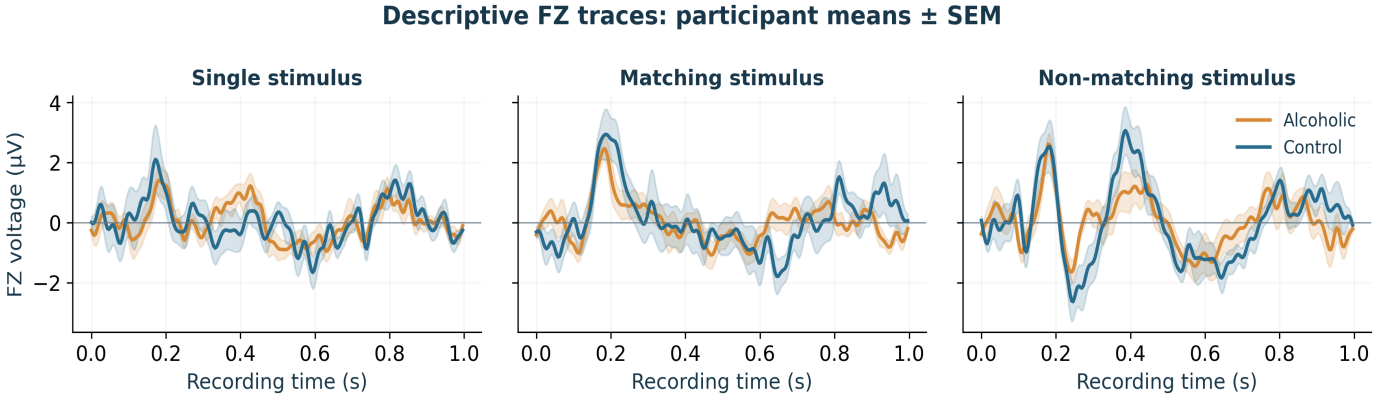
Supplementary EEG visuals

Analysis workflow



The two outputs share preprocessing and participant-separated validation, then use different feature representations and target labels. This figure is a schematic of the implemented workflow, not a brain anatomy illustration.

Real FZ recording-relative waveforms



Each trace averages trials within a participant first, then averages participant means within a group and condition. Shading shows one SEM across ten participant means. All plotted voltages come from the original dataset after the same 1–40 Hz filtering; no waveform was synthesized.

Stimulus onset was not verified in this export. These curves cannot be described as confirmed stimulus-locked ERP responses, nor does separation of two mean curves establish a group effect. The standalone SVG/PNG files offer a larger, editable view for a poster.

Reading the plotted evidence

Use the accuracy and confusion figures to report model performance. Use this waveform figure as an illustration of the measured signal and averaging procedure. Keep the descriptive group-accuracy graph separate from claims about physiological differences.